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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

December 2024

MA1002-1 – CALCULUS AND DIFFERENTIAL EQUATIONS
(FOR AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- If $r = a\theta$ and ϕ denote the angle between the radius vector and tangent, then $\tan \phi =$
A) $\frac{1}{\theta}$ B) θ
C) r D) $\frac{a}{\theta}$
- The radius of curvature of the curve $x = at^2, y = 2at$ at the point $t = 0$ is
A) $2a$ B) a
C) 2 D) $\frac{a}{2}$
- The function $f(x) = x^2$ satisfy the Rolle's theorem in which of the following intervals:
A) $[1, 2]$ B) $[0, 1]$
C) $[-1, 1]$ D) $[-2, 1]$
- The derivative of arc, $\frac{ds}{d\theta}$, for the curve $r = f(\theta)$ is :
A) $\sqrt{r^2 + \left(\frac{d\theta}{dr}\right)^2}$ B) $\sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}$
C) $\sqrt{1 + r^2 \left(\frac{d\theta}{dr}\right)^2}$ D) $\sqrt{1 + r^2 \left(\frac{dr}{d\theta}\right)^2}$
- Given $u = x + y, v = xy$ then the Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ is :
A) $x + y$ B) 0
C) $x - y$ D) 1
- If $z = 6x^2y^2 + 8x^2$ then $\frac{\partial z}{\partial x}$ is :
A) $6y^2 + 16x$ B) $6x^2 + 8x^2$
C) $12xy^2 + 16x$ D) $6x^2 + 6y^2 + 8x^2$
- If $f(x)$ is a function such that $f_{xx} = 6xy^3$ then $f_{xyx} =$
A) $3xy^2$ B) $3x^2y^3$
C) $6y^3$ D) $18xy^2$
- If $f(x, y) = 2x^3 - 4y^2$, what is the value of f_x and f_y at $(3, 2)$
A) $f_x = 54, f_y = -16$ B) $f_x = 16, f_y = -54$
C) $f_x = 54, f_y = 0$ D) $f_x = -54, f_y = -16$
- If $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ then $\text{div}(\vec{r})$ is :
A) 1 B) 3
C) $\hat{i} + \hat{j} + \hat{k}$ D) 0

10. If $f(x, y) = 5xy + 2x^2$ then Hessian of f is
 A) $\begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$ B) $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$
 C) $\begin{bmatrix} 4 & 5 \\ 5 & 0 \end{bmatrix}$ D) $\begin{bmatrix} 4 & 1 \\ 1 & 0 \end{bmatrix}$
11. If $\vec{F} = ax\hat{i} - by\hat{j}$ is solenoidal then which of the following is true?
 A) $a = b$ B) $a^2 - b^2 = 0$
 C) $a + b = 0$ D) $a^2 + b^2 = 0$
12. If $\vec{F} = 6\hat{i} + \hat{j} + by\hat{k}$ then $\nabla \times \vec{F} = \vec{0}$ if.
 A) $b = 2$ B) $b = -1$
 C) $b = 0$ D) $b = 1$
13. Which of the following function is a solution of the differential equation $\frac{dy}{dx} = y$?
 A) $y = 3e^{2x}$ B) $y = 2e^x$
 C) $y = e^{3x}$ D) $y = e^{-x}$
14. If the roots of the auxiliary equation of a differential equation are 0, 1, -1 then the differential equation is:
 A) $D^3y = 0$ B) $(D^3 - D^2)y = 0$
 C) $(D^3 - 1)y = 0$ D) $(D^3 - D)y = 0$
15. The partial differential equation of the expression $z = e^y g(x)$, where $g(x)$ an arbitrary function, is:
 A) $z = \frac{\partial z}{\partial y}$ B) $z = \frac{\partial z}{\partial x}$
 C) $\frac{\partial z}{\partial x} = 0$ D) $\frac{\partial z}{\partial y} = 0$
16. Which of the following equation is not a hyperbolic partial differential equation?
 A) $-2 \frac{\partial^2 u}{\partial x^2} - x^2 \frac{\partial^2 u}{\partial y^2} = 0$ B) $\frac{\partial^2 u}{\partial x^2} - \frac{\partial^2 u}{\partial y^2} = 0$
 C) $xy \frac{\partial^2 u}{\partial x^2} - 3xy \frac{\partial^2 u}{\partial y^2} = 0$ D) $\frac{\partial^2 u}{\partial x^2} - y^2 \frac{\partial^2 u}{\partial y^2} = 0$
17. The value of $\int_0^2 \int_0^y x \, dx \, dy$ is equal to
 A) $4/5$ B) $2/3$
 C) $4/3$ D) $1/5$
18. The value of integral $\iint_D dx dy$ where D is the region above the x -axis and within a circle centered at the origin of radius 2.
 A) 4π B) 8π
 C) $\frac{5}{4}\pi$ D) 2π
19. The volume of the solid bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 1$ is given by:
 A) $\int_0^1 \int_0^{1-x} \int_0^{1-x-y} dz \, dy \, dx$ B) $\int_0^1 \int_0^x \int_0^{x+y} dz \, dy \, dx$
 C) $\int_0^1 \int_0^{1-x} \int_0^{1-y} dz \, dy \, dx$ D) $\int_0^1 \int_0^x \int_0^{x-y} dz \, dy \, dx$
20. The value of $\beta(3, 2)$ is:
 A) $1/12$ B) $1/2$
 C) $1/10$ D) $1/24$

PART – B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	With usual notation prove that $\tan \phi = r \frac{d\theta}{dr}$.	6	L*1	1	2
b)	Show that the radius of curvature at any point of the cardioid $a(1 - \cos \theta)$ varies as $\sqrt{r}$.	6	L2	1	1

- c) If $x = r \cos \theta$ and $y = r \sin \theta$, find $J = \frac{\partial(x,y)}{\partial(r,\theta)}$. 4 L1 2 1
2. a) A rectangular box open at the top is to have volume of 32 cubic ft. Find the dimensions of the box requiring least material for its construction. 6 L2 2 2
- b) State and prove the Cauchy's Mean value theorem. 6 L1 1 1
- c) If $u = \frac{y}{x} + \frac{z}{x} + \frac{x}{y}$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$. 4 L1 2 1
3. a) If $u = f(y - z, z - x, x - y)$ show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. 6 L2 2 2
- b) Find the angle of intersection between the curves $r = \frac{a\theta}{1+\theta}$, and $r = \frac{a}{1+\theta^2}$. 6 L1 1 1
- c) Obtain the Taylor's Series expansion of the function $\sin x$ in powers of $(x - \frac{\pi}{2})$. Expand up to third degree terms. 4 L2 1 2

Unit – II

4. a) If $\phi = x^3 + x^2y^2 + yz^3 - xy + yz$, find $\vec{F} = \nabla\phi$. Hence find $\nabla \cdot \vec{F}$ and $\nabla \times \vec{F}$. Evaluate these at the point $(1, -1, 1)$. 6 L1 3 1
- b) Solve the differential equation: $(D^3 + D)y = 1 + x^2 - 2x^3$ 6 L2 4 1
- c) Define irrotational vector field. Show that the vector $2xy\hat{i} + (x^2 + 2yz)\hat{j} + (y^2 + 1)\hat{k}$ is irrotational. 4 L1 3 2
5. a) Find the directional derivative of $f(x, y, z) = xy^2 + yz^3$ at the point $(2, -1, 1)$ in the direction of $\hat{i} + 2\hat{j} + 2\hat{k}$. 6 L2 3 2
- b) Solve the differential equation: $(D^3 + 6D^2 + 9D)y = e^{-3x} - \sin 3x$ 6 L2 4 2
- c) Form partial differential equations by eliminating the arbitrary constants/or function.
(i) $z = ax^2 + by^2$
(ii) $z = f(x^2 + y^2)$ 4 L2 4 1
6. a) Evaluate the gradient and Hessian of the function $f(x, y) = 2xy^2 - 3x + 4yz^3$ at the point $(1, -1, 1)$. 6 L1 3 1
- b) Solve the P.D.E. by the method of separation of variables:
 $u_x = 2u_t + u$. 6 L1 4 2
- c) Find the particular solution of the differential equation: $y'' - 2y' = 0$, given that $y(0) = -3$ & $y'(0) = 2$. 4 L1 4 1

Unit – III

7. a) Show that the area between the parabolas $y^2 = 4ax$ and $x^2 = 4ay$ is $\frac{16}{3}a^2$. 6 L1 5 1
- b) Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ 6 L2 5 2
- c) Evaluate the triple integral: $\int_0^1 \int_0^2 \int_1^2 (x + y + z) dx dy dz$. 4 L1 5 2
8. a) Prove that $\int_0^{\pi/2} \sqrt{\sin \theta} d\theta \int_0^{\pi/2} \frac{d\theta}{\sqrt{\sin \theta}} = \pi$. 6 L1 5 1
- b) Evaluate $\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy$ by changing to polar co-ordinates. 6 L2 5 1
- c) Evaluate integral $\int_0^\infty x^6 e^{-2x} dx$ in terms of Gamma functions 4 L1 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
